

STUDENT LABORATORY MANUAL

Module 3: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing Companion:

Module 3 Student Guide

Toolchain: VS Code + MSYS2 UCRT64 GCC + FreeGLUT

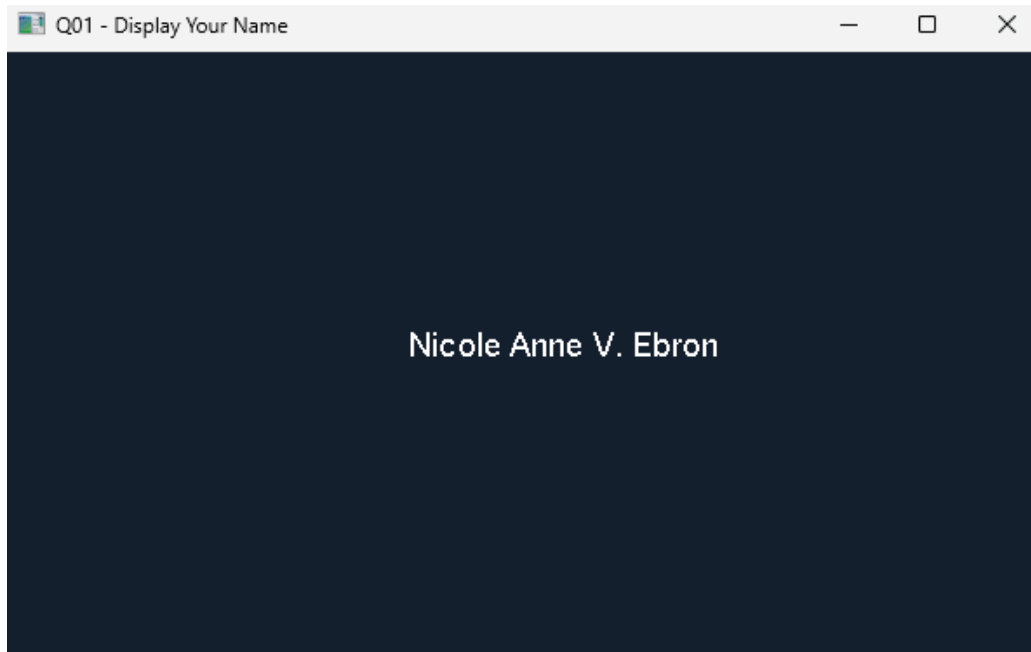
Student Name:	Ebron, Nicole Anne V.
Section:	TN35
Professor:	Dr. Hadji J. Tejuco

GitHub Repository: <https://github.com/nicoleanneebtron-art/Ebron-Nicole-Anne-V>.

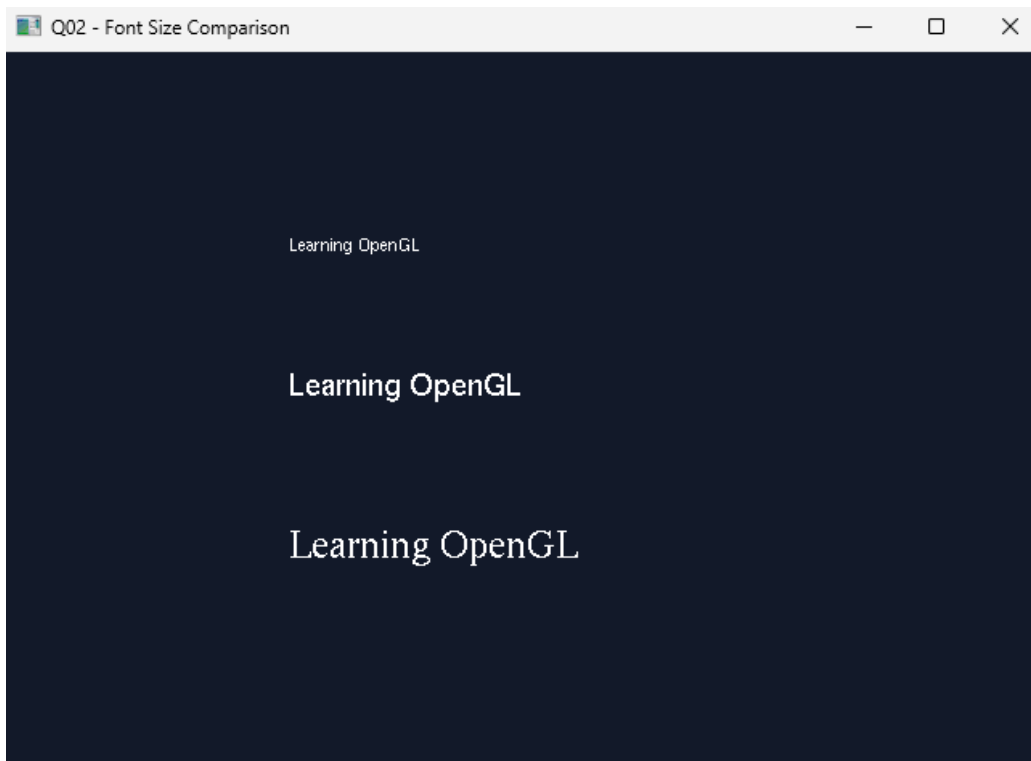
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

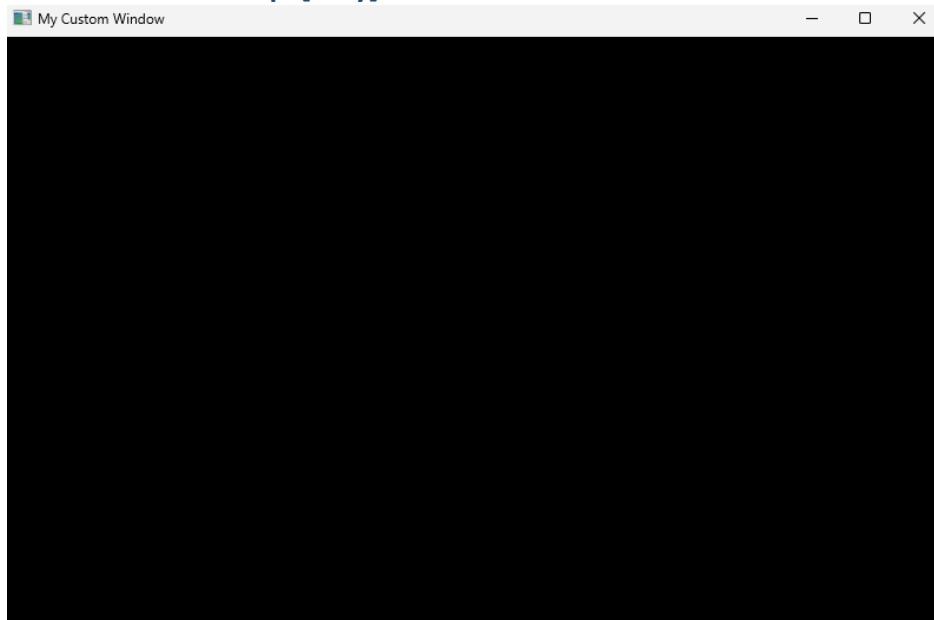
Exercise Q01 — Display Your Name [Easy]



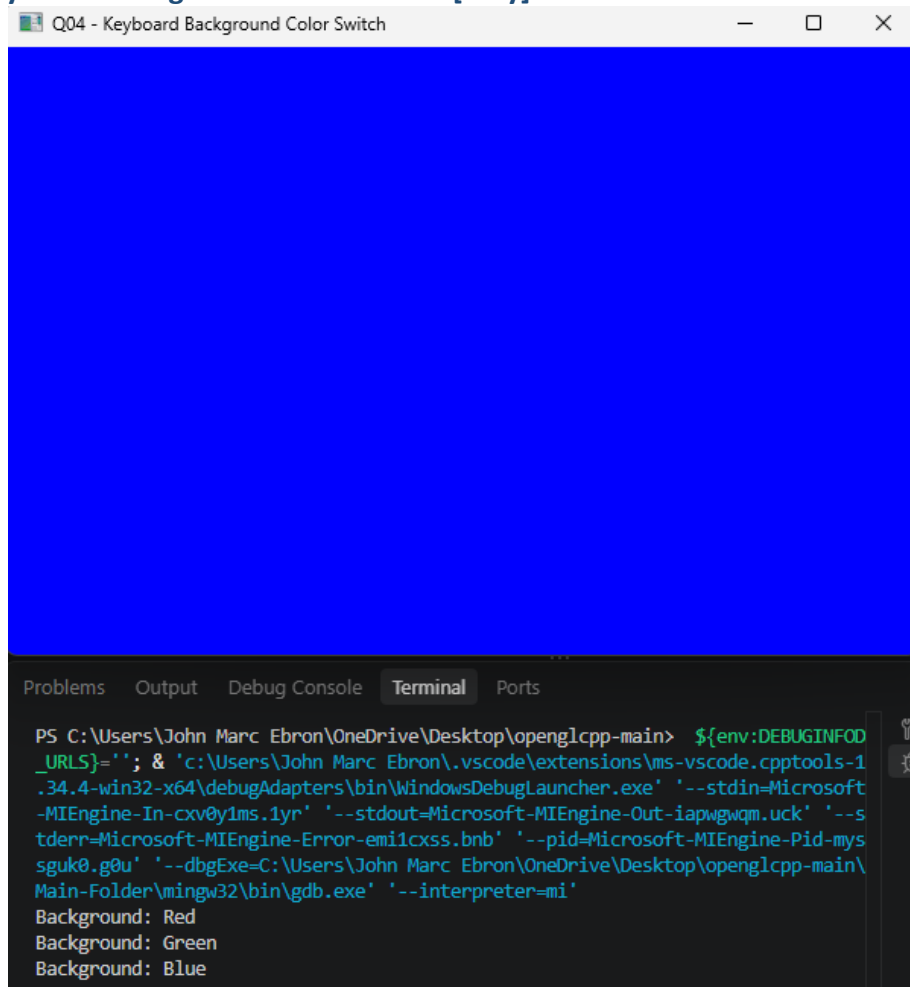
Exercise Q02 — Font Size Comparison [Easy]



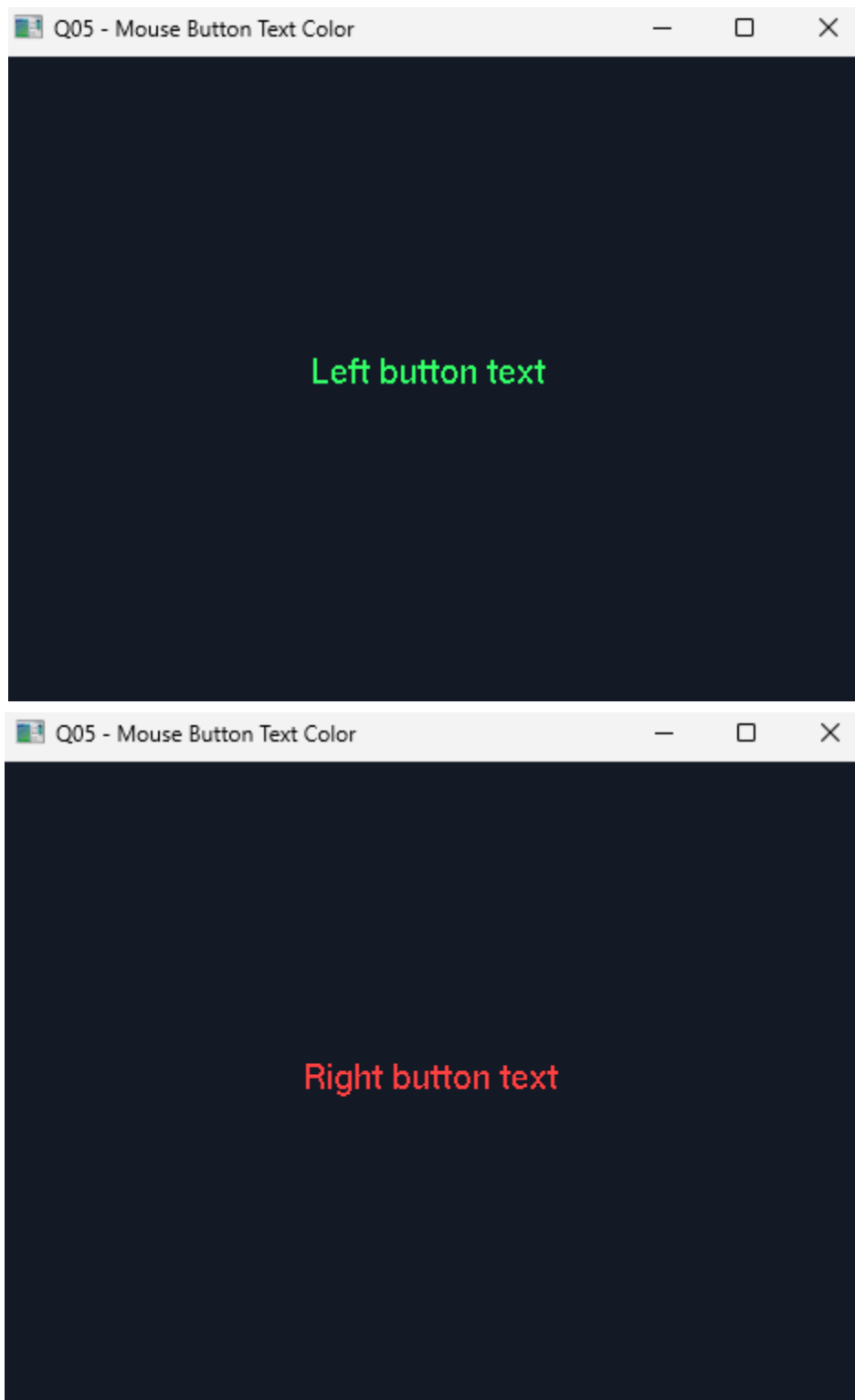
Exercise Q03 — Custom Window Setup [Easy]



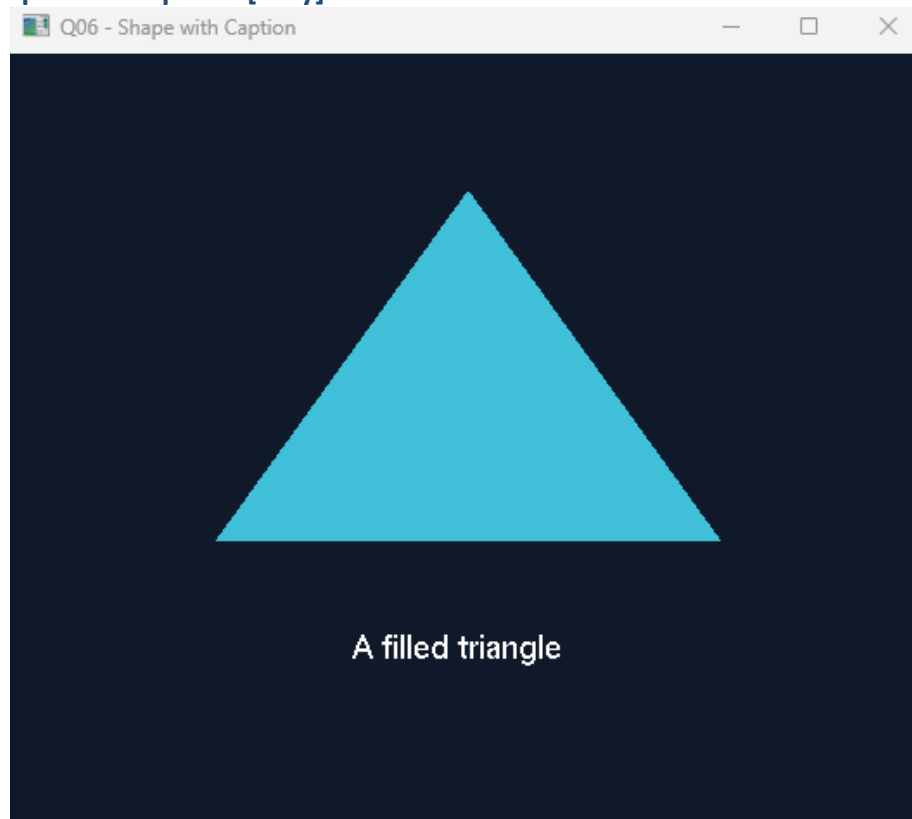
Exercise Q04 — Keyboard Background Color Switch [Easy]



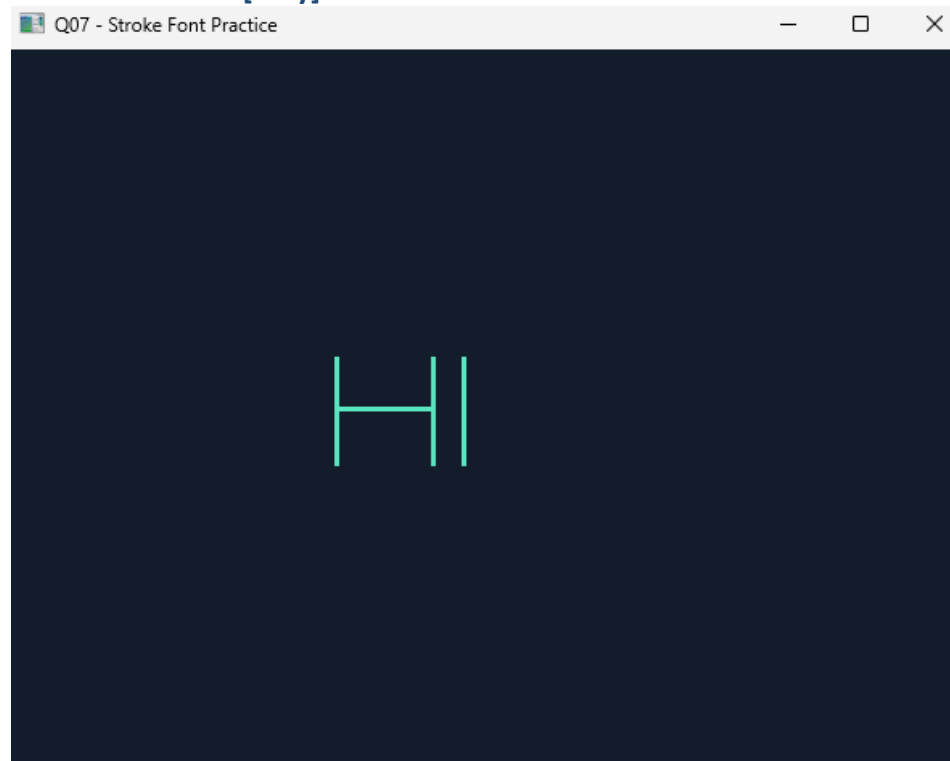
Exercise Q05 — Mouse Button Text Color [Easy]



Exercise Q06 — Shape with Caption [Easy]

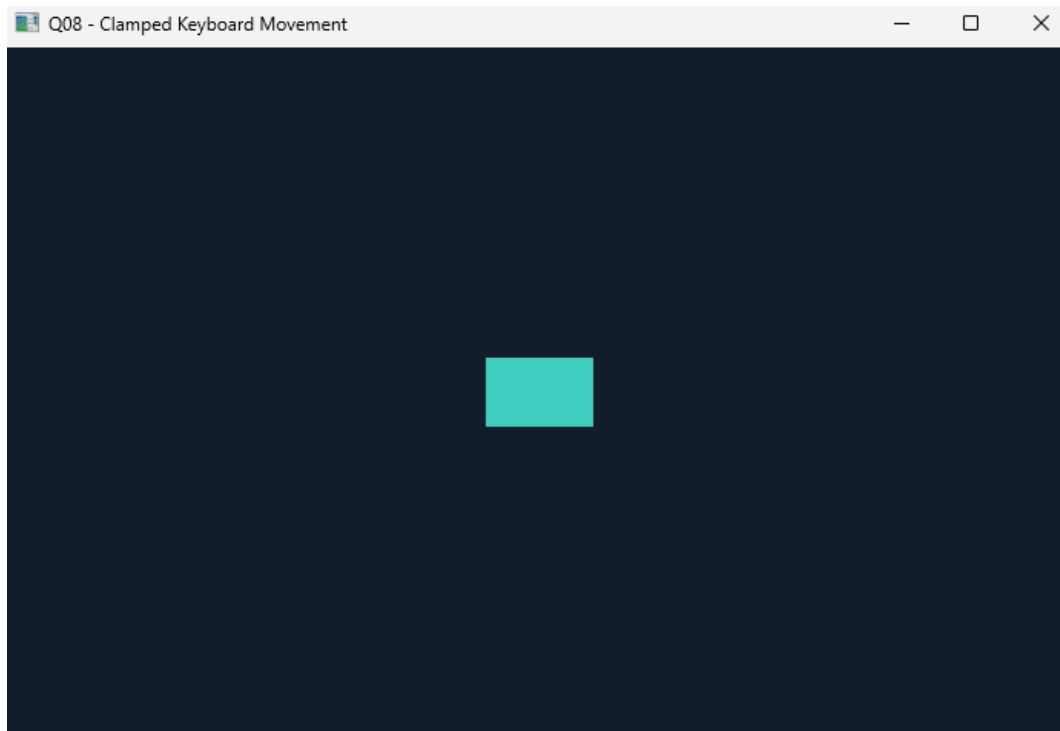


Exercise Q07 — Stroke Font Practice [Easy]

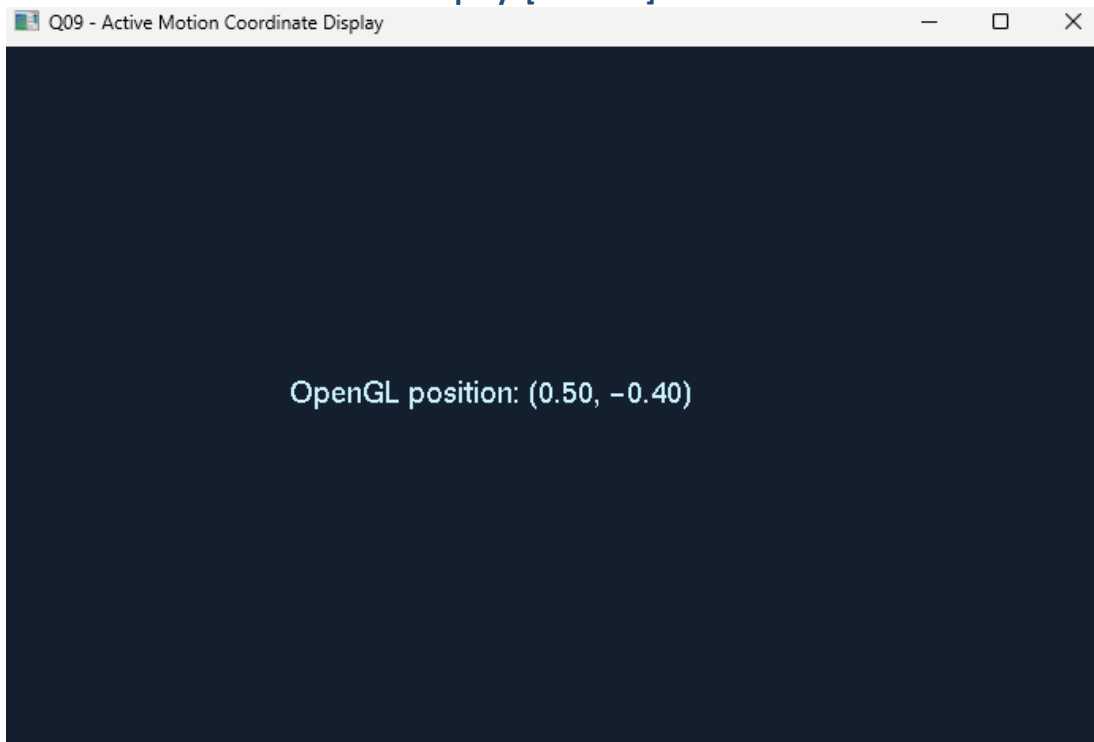


Medium Exercises

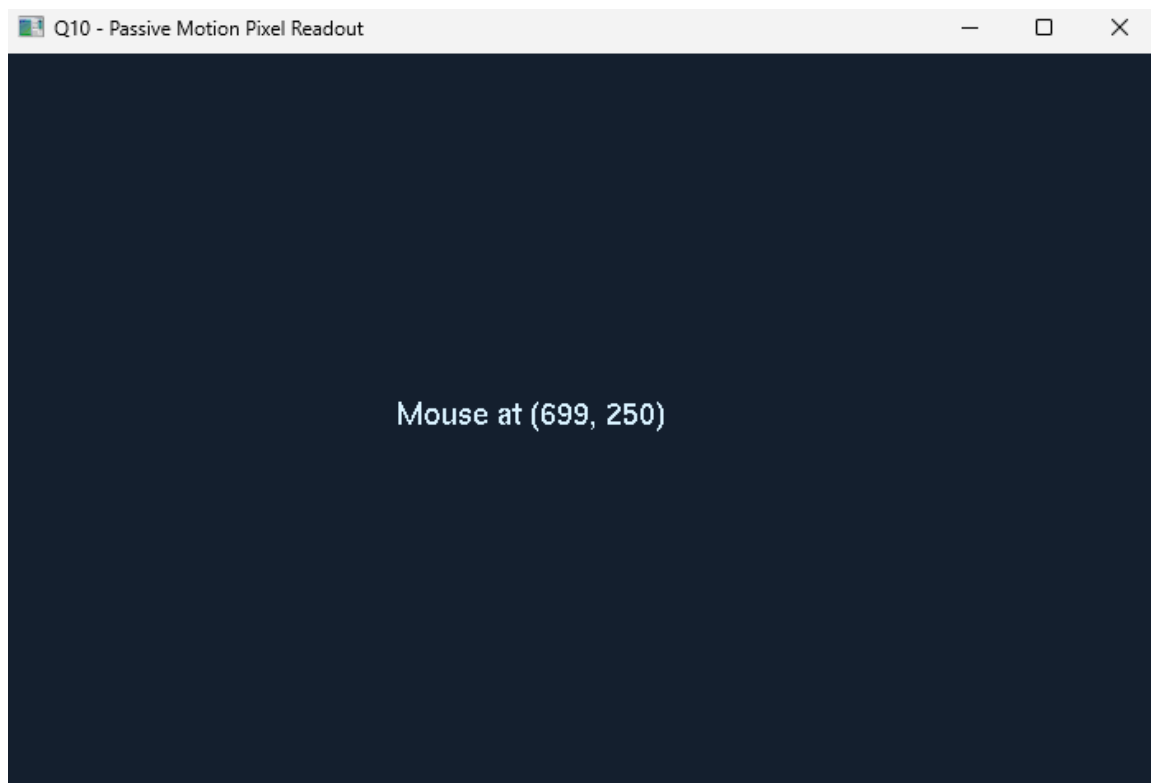
Exercise Q08 — Clamped Keyboard Movement [Medium]



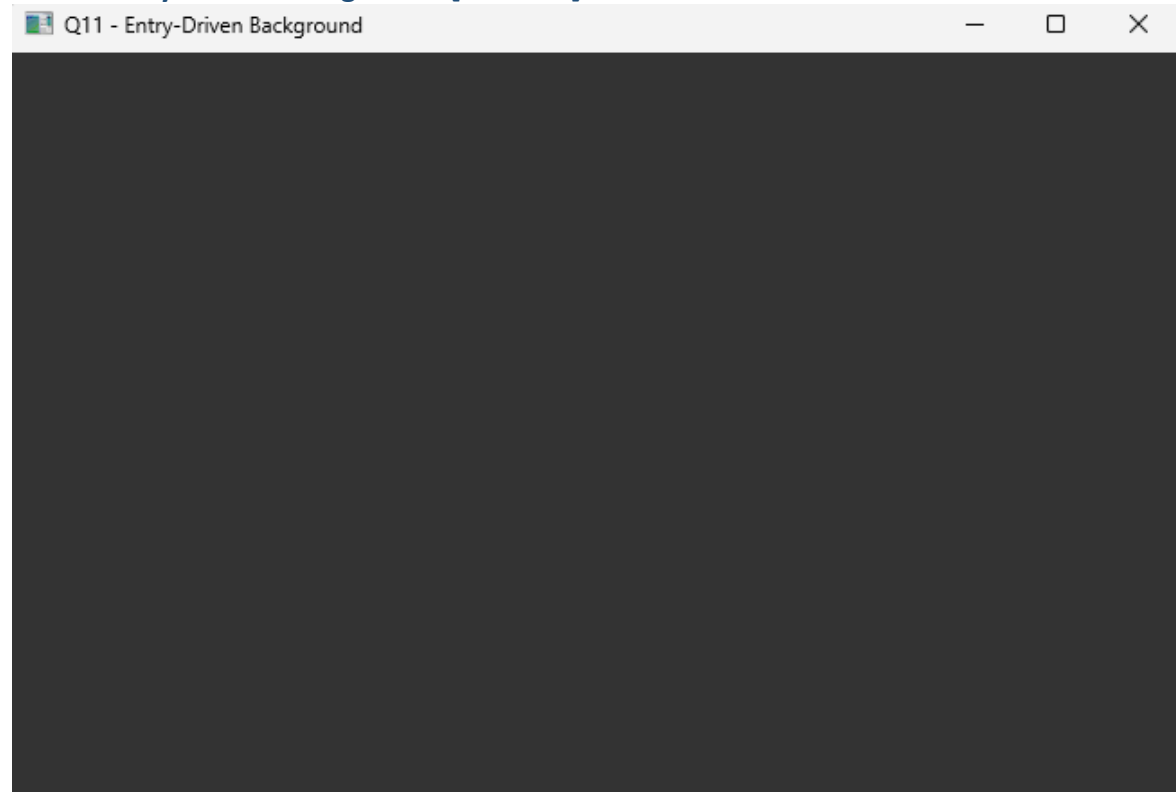
Exercise Q09 — Active Motion Coordinate Display [Medium]



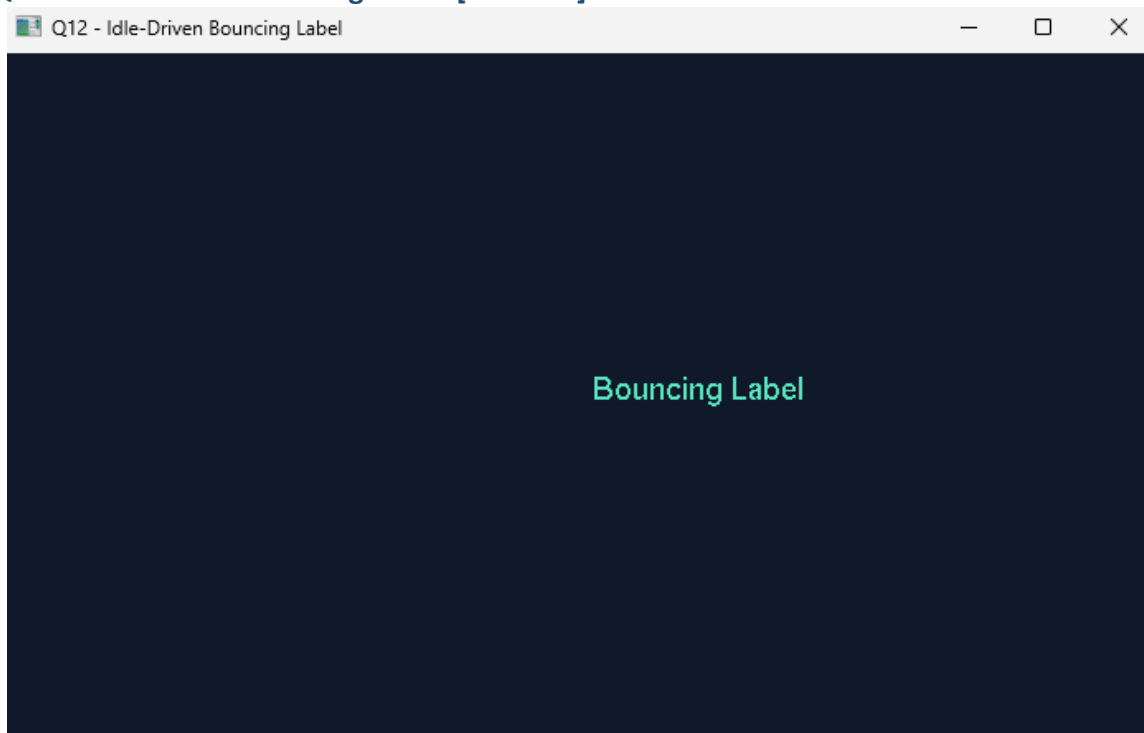
Exercise Q10 — Passive Motion Pixel Readout [Medium]



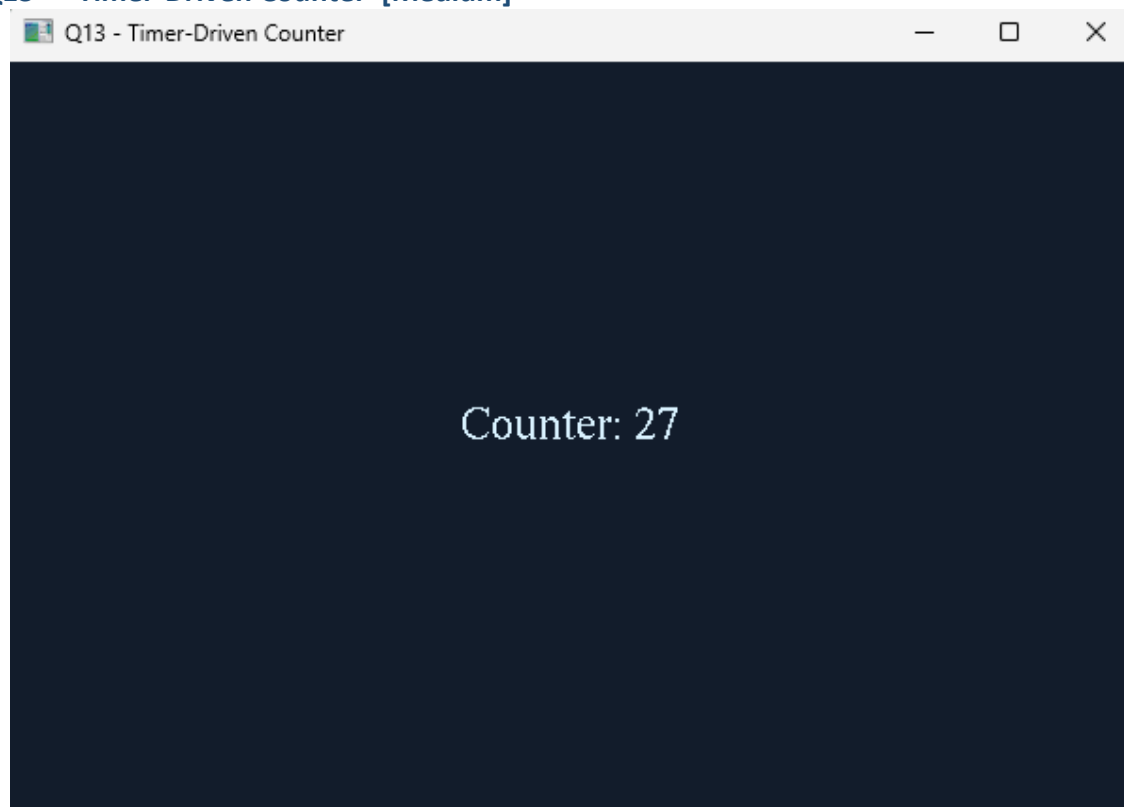
Exercise Q11 — Entry-Driven Background [Medium]



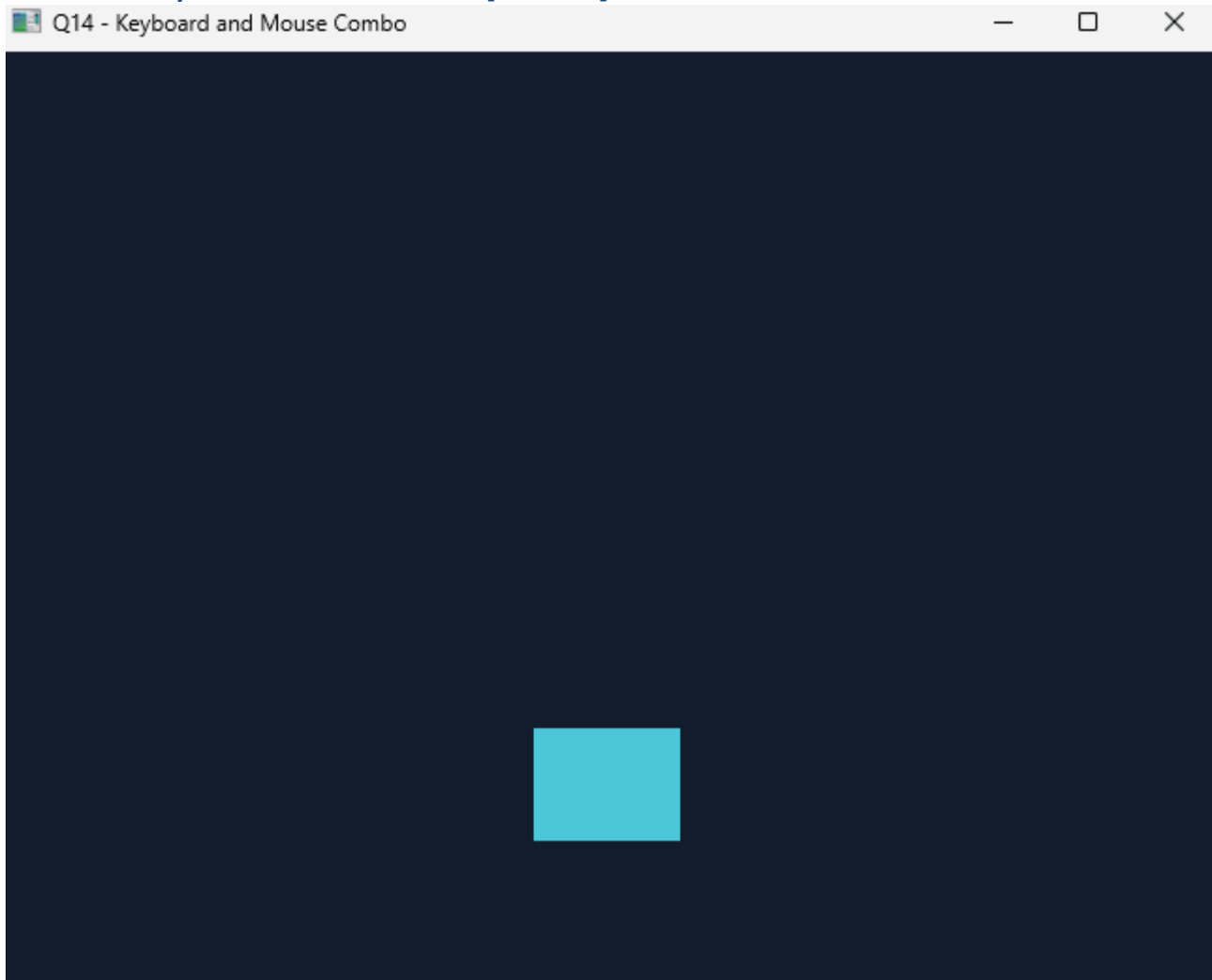
Exercise Q12 — Idle-Driven Bouncing Label [Medium]



Exercise Q13 — Timer-Driven Counter [Medium]



Exercise Q14 — Keyboard + Mouse Combo [Medium]

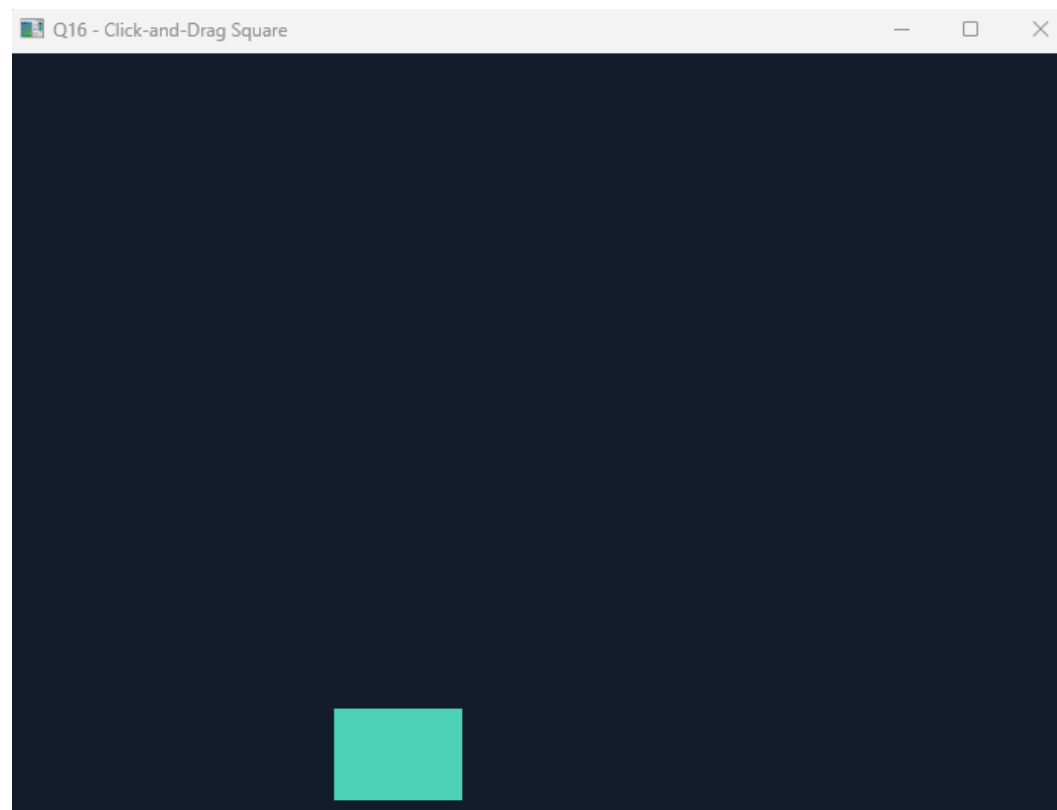
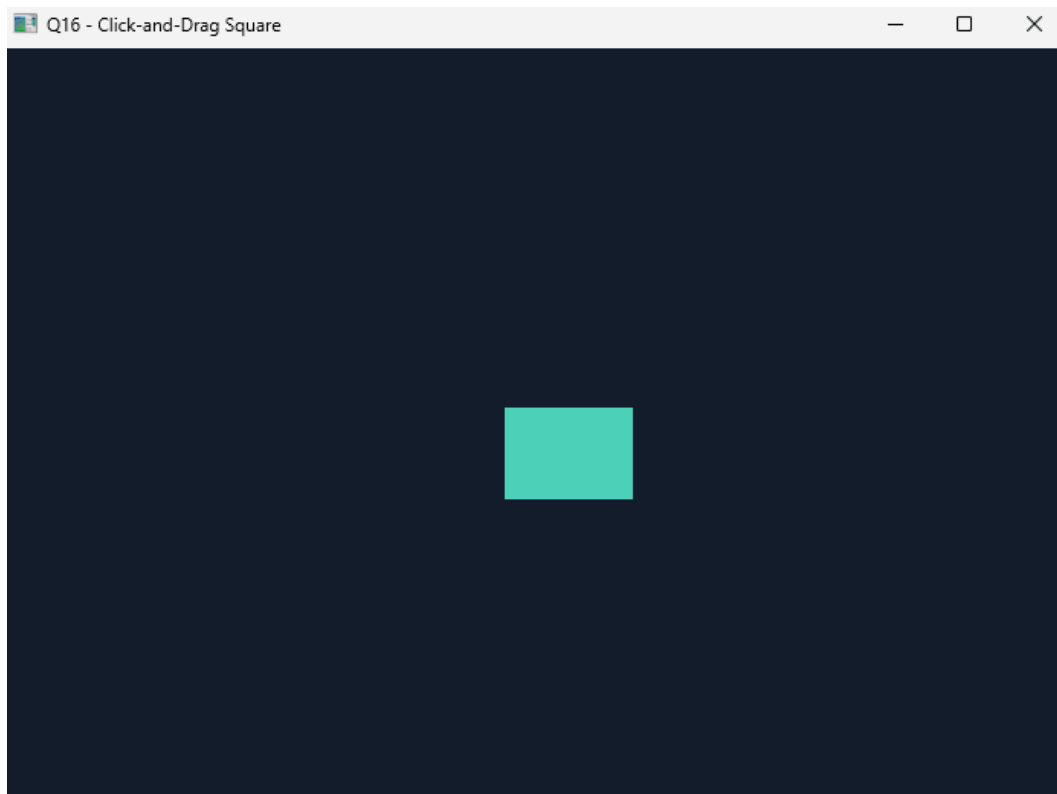


Hard Exercises

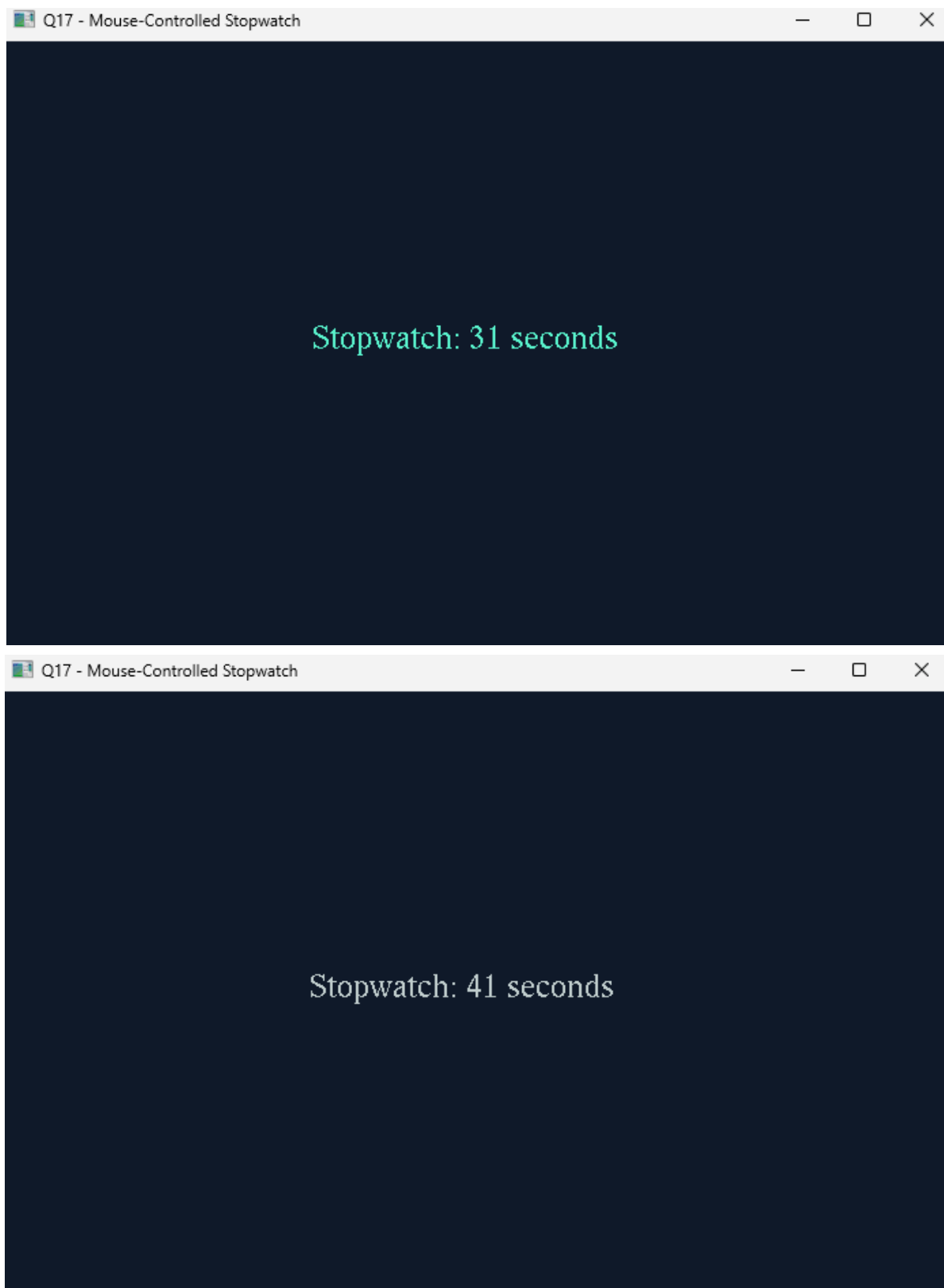
Exercise Q15 — 30-Second Countdown Timer [Hard]



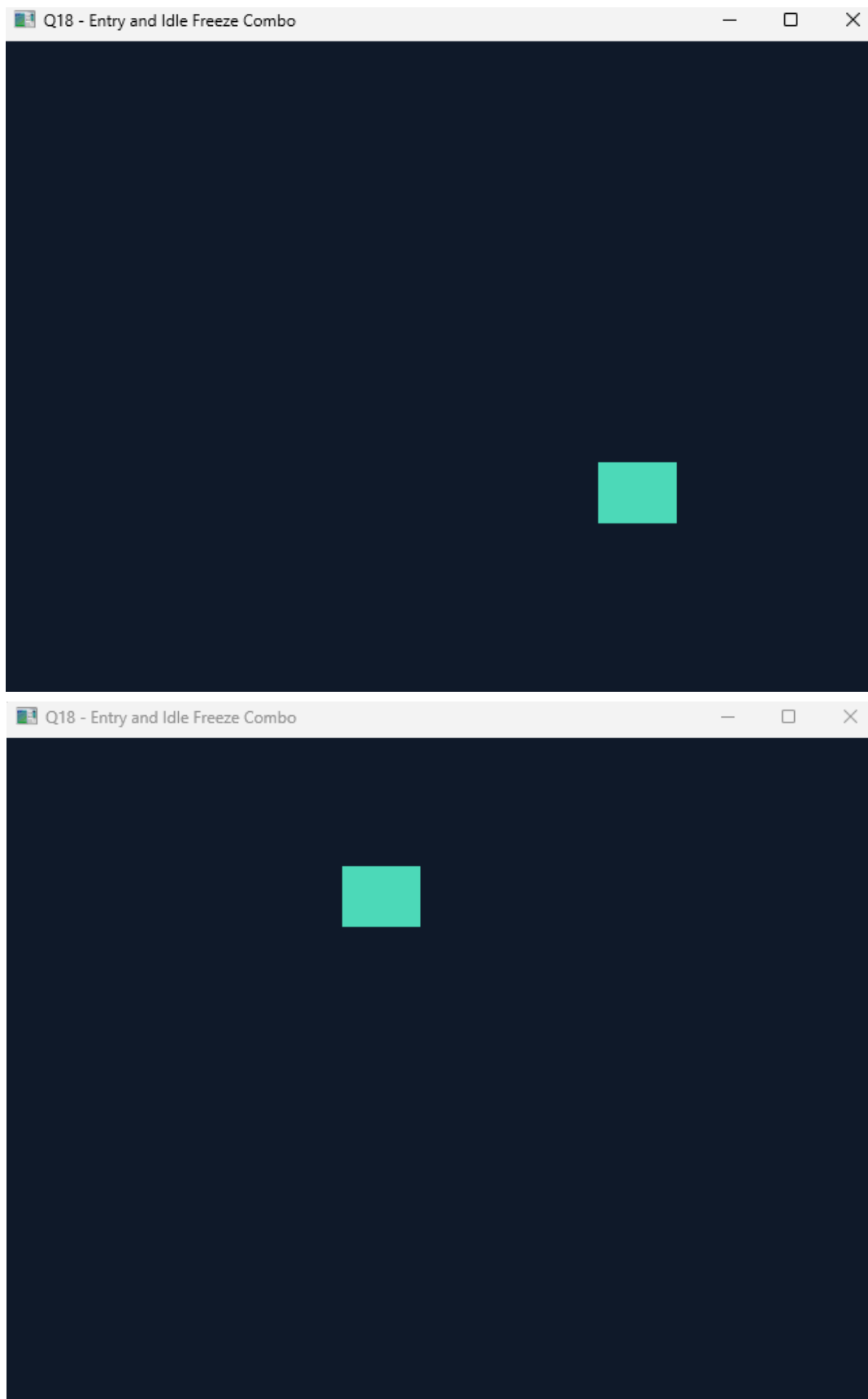
Exercise Q16 — Click-and-Drag Square [Hard]



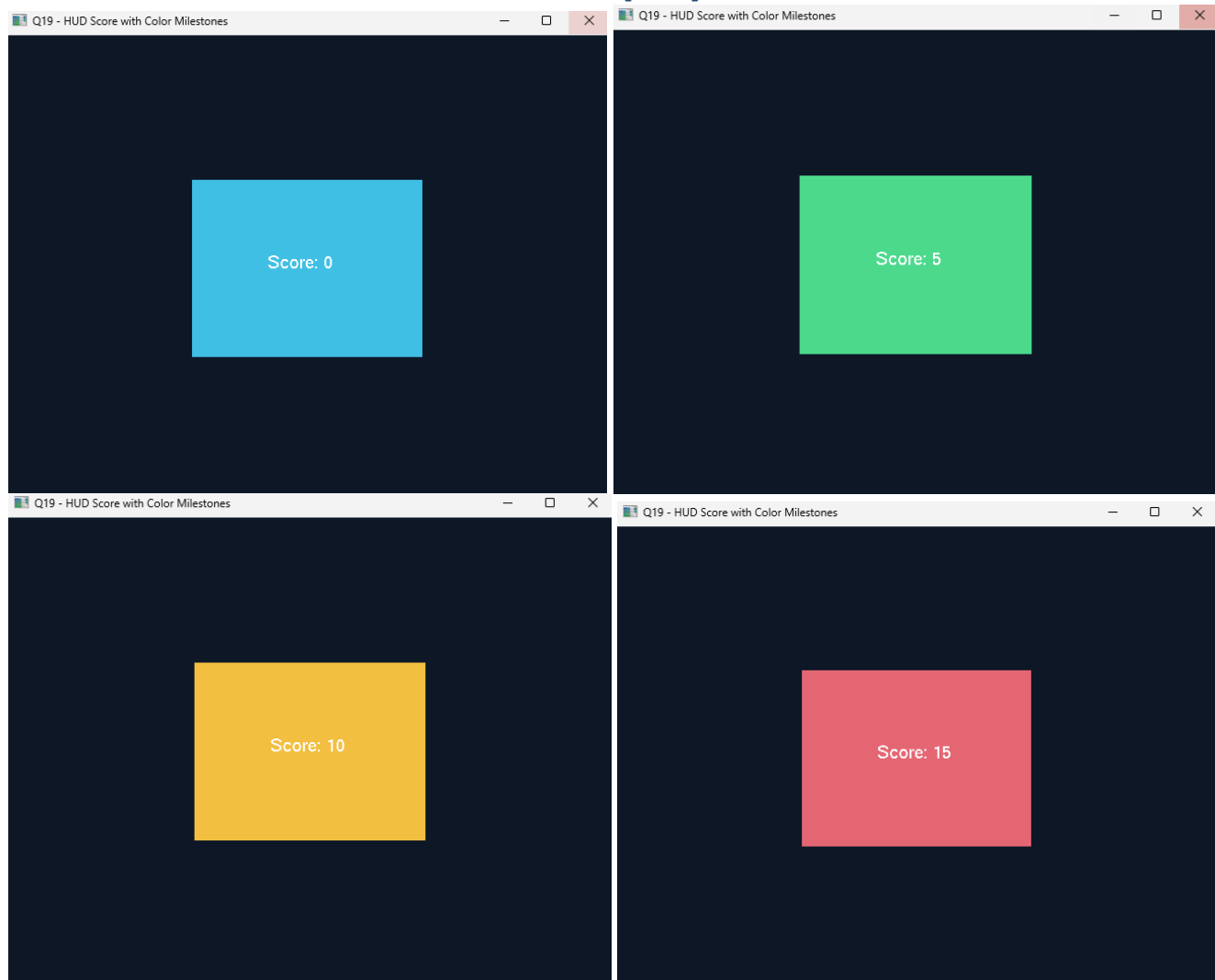
Exercise Q17 — Mouse-Controlled Stopwatch [Hard]



Exercise Q18 — Entry + Idle Freeze Combo [Hard]



Exercise Q19 — HUD Score with Color Milestones [Hard]



Exercise Q20 — Interactive Text Placer (Capstone) [Hard]

